

Title

Microbiota-based interventions for treatment-resistant bipolar depression: study protocol

Abstract

Rationale: Emerging evidence links dysbiosis to mood disorders. This pilot trial will examine whether targeted microbiota manipulation can enhance standard psychiatric care in treatment-refractory bipolar depression. **Methods:** Adults with confirmed bipolar disorder (I or II) presenting with moderate-to-severe depressive symptoms despite ongoing medication will be enrolled (n=35). Eligible participants must be on stable psychiatric medication for ≥ 4 weeks. Participants will be randomly allocated (1:1) to receive either donor-sourced microbiota transplantation or self-microbiota transplantation (autologous), delivered via colonoscopy. Both groups will continue usual psychiatric treatment. **Blinding:** evaluators and participants will be masked to treatment allocation; microbiota composition will be characterized using sequencing. Follow-up assessments occur weekly for the first month, then monthly through week 24. **Primary outcome:** depressive symptom severity assessed via validated depression rating scale. **Secondary:** anxiety, functional capacity, tolerability, and microbiota diversity changes. **Study design:** double-blind, parallel-arm pilot randomized controlled trial.